FUNCTIONAL REQUIREMENTS

The following functional requirements specify the expected behaviour of the Healthcare Management System. These requirements describe the system's functionality in terms of how users will interact with it and how it will manage data.

Requirement 1: User Management

User Registration and login

The system will provide functionality for users to create accounts with unique usernames and passwords. Users will be able to login to access the system and their associated data. The system must enforce password complexity rules to ensure security.

User roles and permissions

The system will support multiple user roles, such as Basic users, advanced users and system administrators. Basic users should have limited access to core operations. System administrators will be able to create, edit, and delete user accounts and assign appropriate roles and permissions

Requirement 2: Doctor/Nurse profile and client operations Management

Addition and viewing of profiles and book

The system will allow nurses/doctors to add their profiles and the clients can be able to view them.

The clients will also be able to add/book an appointment with the doctor/nurse of his/her choice

Updating Records

Users will be able to update the information according to their areas. For example, doctors and nurses will have the ability to update/edit their profiles and also the clients can also be able to update the details of their bookings

Deletion of records

The system will allow users with appropriate permissions to delete records

System features

System features are system components that will be used to meet the identified requirements or provide major services specified by the user. The system features used in this healthcare management system are database, login screen, sign up screen, home screen, nurses and doctor's section, dashboard screen(for client and doctors), appointment screen, about screen, admin section.

Database (High priority)

A database is an organized collection of structured information or data typically stored electronically in a computer system. In this Healthcare Management System, the database is a high priority, this is because it stores all the data entered and generated. I have to use a relational database in this system for it consists of tables and columns. The system uses MySQL database. Some benefits of this database are: reduced data redundancy, ensures data integrity and improves data security and data access. The cost of this data base is not high

Login Screen/Sign up Screen

A login screen is a user interface that allows a user to authenticate themselves with a computer system by entering their login credentials. A sign up screen is used to register a user using login details such as username and passwords which will allow them to login in to the system. This is a medium priority to the system whereby the user can be logged into the system and still use the other system features. The benefit of the login screen/sign up screen is to improve or enforce security measures that will protect the data of the system.

Home screen

The home screen is whereby the doctor's and nurses profiles will be viewed and clients can be able to search the doctor/nurse using filters such as name, specialization and location filters using the search bar that will be on top of the screen

Dashboard Screen

The dashboard screen is whereby the user can be able to:

- View Bookings: Upcoming and completed services.
- Track Service: Real-time updates (e.g., "Nurse en route" or "Service Completed").
- View payment history and download invoices.

About Screen

This screen helps the new users of the system with information about the organization; the foundation, the operations of the organization and its purpose. It also has contact information of the hospital and a user can choose the means which he/she can contact the organization (Phone call, SMS, Email), to inquire more information about the organization.

Appointments Screen

This screen will enable a patient/client to book an appointment with the desired doctor/nurse

It will contain a form which will enable the client to fill in the required information

Admin section

This section is only accessible by the administrators of the system because they have a login section whereby they enter the credentials only known by the administrators. In this section, the administrators can be able to make changes to some sections. The normal users can not be able to view appointments made by other patients. But this is where the admin section comes in because through the admin section, these features can be accessed. This will reduce data redundancy.

Analytics Dashboard: Insights into platform activity, including revenue, active users, and regional demand

Payment monitoring

Screen layout

All screens will follow a consistent layout to ensure easy navigation and a seamless user experience. Each screen will have a content area whereby the main part of the screen will display content specific to the current action.

GUI Standards and guidelines

The user interface will follow a predefined style guide to maintain a consistent look. This guide will define:

- Colors – A Defined color scheme will be used across the system for consistency.
- Fonts – A standard font family and size will be used for all text in the application to ensure readability.

GUI Standards

- Error messaging – Clear and informative error messages will be displayed, guiding the user to correct any users e.g. Error
- Confirmation prompts – Any action that can cause data loss or significant changes will prompt the user to confirm the action e.g. deleting a profile

Error Handling and Notification

The system will display error messages that clearly explain the problem and offer solutions where possible. For example, If the user enters a wrong login credentials, the system will display wrong username or password.

Success notification – After successful completion of action e.g. creating a profile the system will display a confirmation message that is ‘Profile successfully created’

Hardware interfaces

This section describes the logical and physical characteristics of each interface between the Healthcare Management System and the hardware components it interacts with.

Supported device types

- Local and external storage devices – hard drives or external storage for storing data and backing up data.
- Mobile Smart Phones and laptops - the system will run on browsers which are supported by mobile phones and laptops/PCs

Data and control interactions

- Local and external storage devices:
  a. Data interaction – The HMS must read from and write to local storage that is internal hard drives or external storage devices e.g. USB drives and external hard drives for tasks such as adding data, backups and restoring data from backups.
  b. Control interaction – The system will provide options to restore data from backups, verify the integrity of saved data
  c. Communication protocol – For external storage, the HMS will use standard Universal Serial Bus protocols for data transfer.
- Mobile Smart Phone:

a. Data interaction – The HMS will run on browsers supported by mobile smart phones and laptops and interact with the systems memory, storage and input/output devices
b. Control interaction – The software will control CPU resources, memory allocation and storage access areas.
c. Communication protocol – The system will use the operating systems built in communication protocols for interacting with hardware

Software interfaces

A software interface can be described as the mechanism that allows different software components or systems to communicate and exchange data with each other. This section describes the connections between the HMS software and other specific software components, including databases, operating systems, tools, libraries, and integrated third-party or commercial components, detailing the purpose of each interaction, communication protocols and any implementation constraints.

Database interface

Database type and version

The HMS will use a relational database management system (RDBMS) which is MySQL for data storage and retrieval.

Data exchange

The HMS will interact with the database to store and retrieve data including patient details, nurse/ doctor details, appointment details.

Data entered or edited/updated by the user will be sent to the database for storage.

The system will query the database by displaying information on screens such as doctor profiles.

Communication Protocol

The HMS will use MySQL for communication with the database.

Data Sharing

Data will be shared across different modules of the HMS through database queries

Implementation constraint

All database transactions will have principles to ensure data integrity and reliability.

Operating system interface

Supported OS

- Android 5.0 or higher
- Windows 10 or higher
- Linux
- MacOs
- iOS

OS services needed

The system will rely on core operating system services for file management, memory management and process handling

Communication protocol

The HMS will communicate with the OS via standard system calls, using operating system libraries to perform file operations, memory management, and peripheral handling.

Integrated Development Environment (IDE)

This is the tool that is used to develop various systems

In this case, an IDE called Visual Studio Code will be used since it is free unlike Pycharm Professional which requires a paid license for use

Web hosting

At First, the free web hosting plan will be used. Then afterwards other plans might be incorporated

APIs (Application Programming Interfaces)

These are software that are integrated with other softwares for better performance.

In this case, the MPESA API will be integrated with the HMS to facilitate the paying of the doctors/nurses. This will be achieved through Mpesa Daraja.

Potential risks and hazards

Data loss and corruption

- Risk of data loss or corruption due to system failures, power outages or incorrect data entry.
- Safeguard – The system will include backups that occur regularly.

- Prevention – The database will immediately save data when its added and data loss will be prevented.

Unauthorized access

- Risk of unauthorized access to sensitive business information
- Safeguard – The system will implement a robust user authentication. Only authorized users will be allowed to make changes in the system.
- Prevention – The use of login credentials which will prevent unauthorized access

Technology Stack

- **Frontend**: HTML, CSS, JavaScript, Bootstrap
- **Backend**: Django (Python)
- **Database**: MySQL / SQLite (for development)
- **Hosting**: InfinityFree (for initial deployment)

TIME FRAME

Phase	**Task Description**	**Estimated Time Frame**
Phase 1: Requirement Analysis	Collecting and documenting requirements	1 week

Phase 2: UI/UX Design	Creating wireframes and design mockups	2 weeks
Phase 3: Frontend Development	Implementing the frontend interface	4-5 weeks
Phase 4: Backend Development	Developing backend APIs and database design	4-5 weeks
Phase 5: Integration	Integrating frontend with backend	2 weeks
Phase 6: Testing	Conducting unit, integration, and system tests	2 weeks
Phase 7: Deployment	Deploying the application to a hosting server	1 week

Total estimate of the project = 16-18 weeks

Budget and Pricing

Item	**Cost Estimate**
Domain Name Registration	Ksh.2000 – Ksh.3000 (per year)
SSL Certificate (Optional)	Free (Let's Encrypt)
Hosting (InfinityFree)	Free
Bootstrap Template	Free / Ksh.2000 – Ksh.5000 (if premium)

Developer Cost (if outsourced)	Ksh.80000-100000
Maintenance	Ksh. 7500 (per month)
Miscellaneous	Ksh.5000 – Ksh.10000